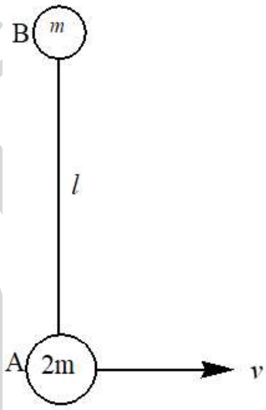
TOPIC : Centre of Mass & Collisions

SUBJECT: PHYSICS **DATE: 20-03-2024**

SINGLE CORRECT TYPE (+3, –1)

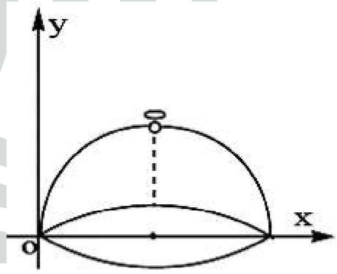
1. Two masses A and B connected with an inextensible string of length l lie on a smooth horizontal plane. A is given a velocity of v m/s along the ground perpendicular to line AB as shown in Fig .Find the tension in string during their subsequent motion.



(A) $\frac{2mv^2}{3l}$ (B) $\frac{3mv^2}{2l}$

(C) $\frac{2mv^2}{l}$ (D) $\frac{mv^2}{3l}$

2. A hollow hemisphere of radius R and mass m is resting on a rough horizontal surface The hemisphere has a small hole at the top. An ant of mass m, moving near the top of sphere falls down into the hole suddenly at t = 0, and reaches the ground in time t_0 and comes to rest. The y co-ordinate of centre of mass of ant and hemisphere is given in column II corresponding to different times given in Column I.



Column - I	**Column - II**
A. at t = 0	p. R/4
B. at $t = t_0$	q. 5R/8
C. at $t = \frac{t_0}{\sqrt{2}}$	r. R/2
D. at $t = \frac{t_0}{2}$	s. 3R/4

(A) A-s, B-p, C-r, D-q (B) A-p, B-s, C-q, D-r (C) A-r, B-p, C-q, D-s (D) A-s, B-r, C-q, D-pq

Passage

A uniform bar of length 12 L and mass 48 m is placed on a smooth horizontal table as shown in figure. A small moth (an insect) of mass 8 m is sitting on end A of the rod and a spider (an insect) of mass 16m is sitting on the other end B. Both the insects moving towards each other along the rod with moth moving at speed 2 v and the spider at half this speed (absolute). They meet at a point P on the rod and the spider eats the moth. After this the spider moves with a velocity $\frac{v}{2}$ relative to the rod towards the end A. The spider takes negligible time in eating the other insect. Also, let $v = \frac{L}{T}$ where T is a constant having value 4.

3. After starting from end B of the rod the spider reaches the end A at a time

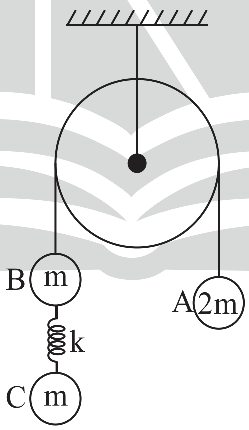
(A) 40 s (B) 30 s (C) 80 s (D) 100 s

4. By what distance the centre of rod shifts during this time?

(A) $\frac{8L}{3}$ (B) $\frac{4L}{3}$ (C) L (D) $\frac{L}{3}$

ONE OR MORE THAN ONE OPTION(S) CORRECT: (+4, –1)

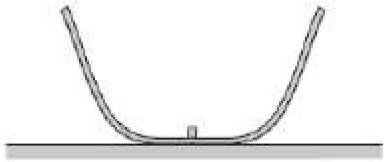
5. Particles A and B are connected by a massless, inextensible cord which goes around the fixed massless pulley. If the system is released from rest with the spring initially unstretched, then:



(A) The maximum extension in the spring is $\frac{2mg}{K}$ (B) The maximum velocity of particle C is $\sqrt{\frac{3m}{4K}}g$

(C) The maximum extension in the spring is $\frac{mg}{K}$ (D) The maximum velocity of particle C is $\sqrt{\frac{3m}{2K}}g$

6. A small block rests on flat bottom of a deep bowl that is placed on a horizontal floor. There is no friction between the block and the bowl as well as between the bowl and the floor. Total mass of the bowl and the block is M. If the bowl is given a horizontal velocity by a sharp hit, the block rises to a maximum height h sliding on the walls of the bowl. Which of the following statements are correct?

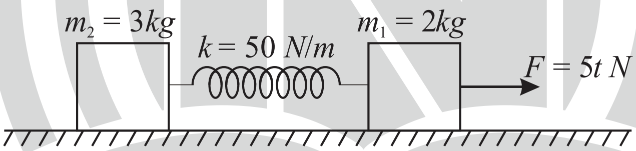


(A) Maximum kinetic energy of the bowl is Mgh.
(B) If the bowl is lighter than the block, minimum kinetic energy of the bowl is zero.
(C) To predict minimum kinetic energy of the bowl, we must know ratio of the masses.
(D) To predict maximum as well as minimum kinetic energy of the bowl, we must know ratio of the masses.

NUMERICALS TYPE: (+4, –1)

7. Two identical small balls A and B each of mass *m* connected by a light inextensible cord of length *l* are placed on a frictionless horizontal floor. With what minimum velocity *u* must the ball *B* be projected vertically upwards so that the ball A leaves the floor? Acceleration of free fall is *g*.

8. On a spring block system shown in figure, a time varying force F = 5 t N is applied on 2 kg mass. After 10s, velocity of 3 kg mass is 30 m/sec, (towards right as shown) Find velocity of 2kg mass at this instant.



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